

The Final Exam

Name
Date
Arithmetic

Instructions

- I) Write a title, name, date, and course.
- II) Number each problem, circle the number, rewrite the problem, and show all work.
- III) Put a rectangle around your answers.
- IV) Write any fractions in your answers in lowest terms and never leave your answers as improper fractions.

① One chef makes $\frac{5}{8}$ of a pound of spaghetti every hour, while another chef makes $\frac{3}{4}$ of a pound of spaghetti every hour. If these chefs work together, how long will it take them to make $\frac{8}{9}$ of a pound of spaghetti (refer to class 35, # SA-D)?

② Jacob unloaded the building material from his truck. The weight of his truck decreased by 5%. If the weight of his truck after unloading was 0.366 tons, what did his truck weigh before he unloaded the building materials? Round your answer to the nearest hundredth.

③ Identify the rule/property demonstrated by each equation.

a) $3 \times \frac{1}{3} = 1$ b) $\frac{7}{8} + \frac{1}{9} = \frac{1}{9} + \frac{7}{8}$ c) $1 \times 5 = 5$ d) $0 + 1 = 1$

④ 2.899 grams of a three gram substance is carbon. What percent of the substance is carbon?

⑤ Convert the rates below to unit rates. Write your answers as mixed numbers.

a) \$26 for 4 lbs of carrots. b) 70 miles for every 9 gallons.

⑥ Draw a bar graph of the following data set. Label the graph properly.

Populations of Various Turtle Species in Rock County

Species	Snapping Turtle	Red-eared Slider	Gopher Tortoise	Painted Turtle
Population	500	700	1,100	200

⑦ a) Write the following number in standard form.

Nine million five hundred fifty-one thousand eight hundred twenty-two

b) Write $<$, $>$, or $=$ between the following two values.

20 cups 5 quarts

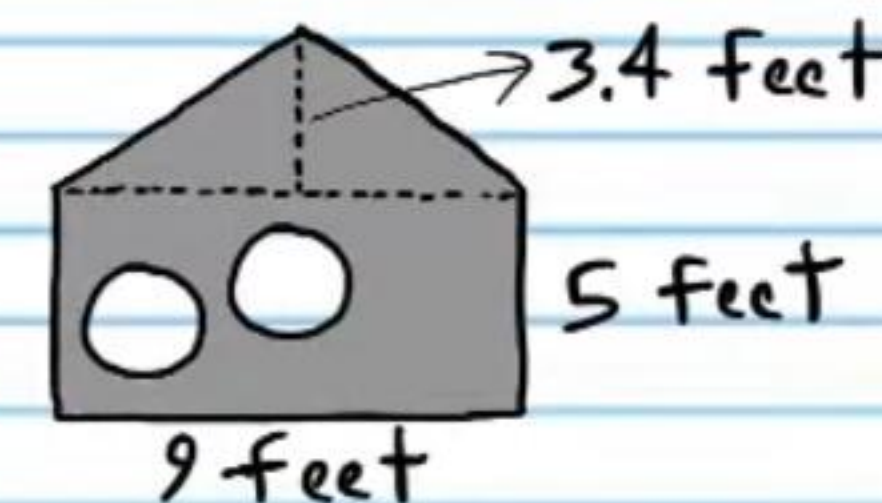
⑧ Arrange the following values from least to greatest.

$0.7, \frac{3}{5}, \frac{5}{8}, \frac{4}{5}$

⑨ Write $<$, $>$, or $=$ between the pairs of values below.

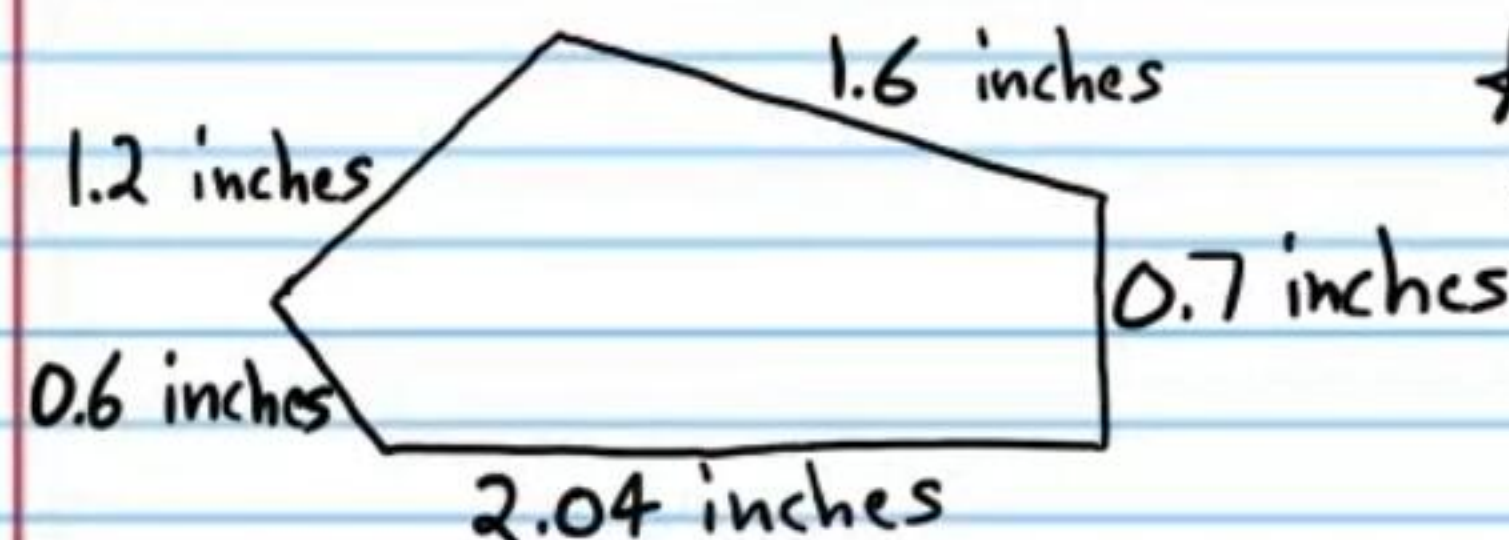
a) $\frac{7}{9}$ $\frac{3}{4}$ b) $\frac{40}{64}$ $\frac{21}{28}$ c) $\frac{63}{81}$ $\frac{2}{3}$ d) $\frac{7}{10}$ $\frac{3}{5}$

⑩ Find the area of the shaded region below if the area of each circle is 3 square feet.



⑪ Justin loans his friend \$6,000 at 11% simple interest paid monthly. If Justin's friend keeps the money for seven months without paying any of it back, how much will he owe Justin in interest? How much will he owe Justin in total?

⑫ Find the perimeter of the figure below.



★ Figure not drawn to scale.

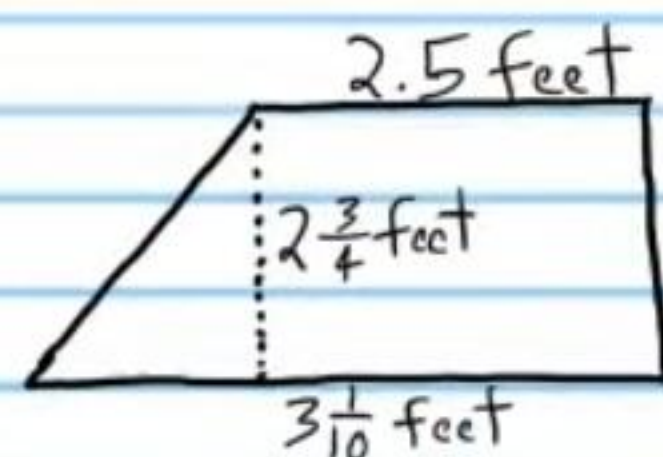
⑬ a) Convert 867.9% to a decimal, then to a mixed number. b) Convert 27.8 to a mixed number, then to a percent.

Beth's Track Times (minutes): 7.18, 7.28, 7.25, 8.08, 7.27, 7.25, 7.07

⑭ Find the mean, median, mode, range, and interquartile range of the data on Beth's track times above. Draw a dot plot of the data and a box and whisker plot above it. Identify any outliers.

⑮ $3\frac{4}{7} - 1\frac{11}{14} + 4 \div 2\frac{1}{3} + 2^3$

⑯ Use the following formula to find the area of the trapezoid below: $A = (b_1 + b_2) \times h \div 2$.



⑰ Alexandria can drive 32.09 miles on one gallon of gasoline. How far can she drive with $7\frac{1}{2}$ gallons of gasoline?

⑱ Thomas does not want to carry more than 10 pounds in his backpack. He has $1\frac{3}{4}$ pounds of beef jerky, $\frac{7}{20}$ of a pound of dried ice cream, 5 pounds of water, and 2.8 pounds of trail mix. Find the total weight of these items. Will Thomas be able to carry all of these items in his backpack?

⑲ Engineers reduced the weight of a racing motorcycle so that it is $7\frac{3}{4}$ kilograms lighter. Before reducing its weight, the motorcycle weighed $134\frac{1}{6}$ kilograms. What is the motorcycle's weight now?

⑳ Cathy had \$8,026.57 in her bank account. She then spent \$6,999.86 for a down payment on a car. How much money is in her bank account now?

㉑ Write a proportion statement and use it to solve the following problem: Lisa plants nine trees every four hours. How many trees will she plant in six hours?

㉒ Jeff wants to fill an empty space in his garden with flowers. If the empty space is 0.8 of a foot by 0.7 of a foot, and it costs \$9.65 to buy enough flowers to fill one square foot of his garden, how much will it cost Jeff to fill the empty space with flowers? Round your answer to the nearest hundredth (i.e. the nearest cent).

23) You and four other people divide a cake equally among each other. One person eats $\frac{4}{5}$ of his share and gives the rest to you, while another person gives $\frac{5}{6}$ of his share to you. What fraction of the entire cake did you get?

24) i) The quotient of a number and five increased by seventeen is 26. Which of the following could be the number?

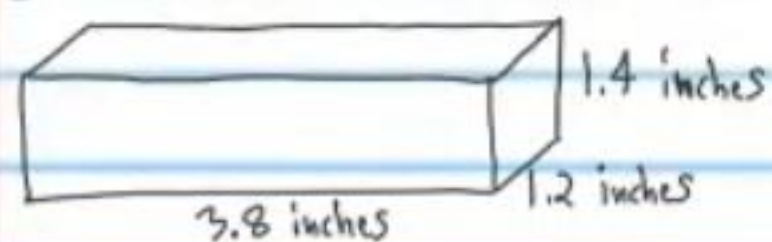
- A 56
- B 45
- C 36
- D 20
- E 65

ii) L is divisible by three. L is also divisible by five. Which of the following could be L ?

- A 10
- B 12
- C 24
- D 30
- E 40

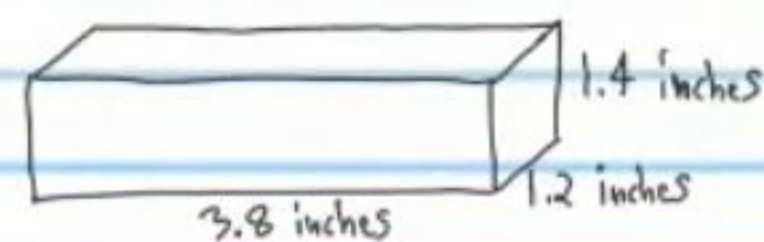
In the following three multiple choice problems, choose the fastest way to find the answers. Write "E" for estimation, "AS" for algorithmic shortcut, or "A" for algorithm.

25A) What is the surface area of the rectangular prism below? (SA = $L \times W \times 2 + L \times H \times 2 + W \times H \times 2$)



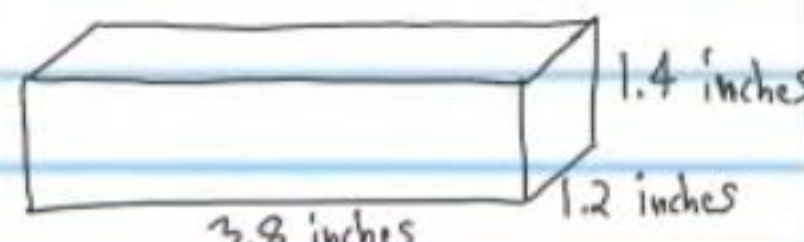
- A 2.312 in²
- B 23.12 in²
- C 0.2312 in²
- D 231.2 in²
- E 2,312 in²

25B) What is the surface area of the rectangular prism below? (SA = $L \times W \times 2 + L \times H \times 2 + W \times H \times 2$)



- A 33.12 in²
- B 23.12 in²
- C 13.12 in²
- D 24.12 in²
- E 25.12 in²

25C) What is the surface area of the rectangular prism below? (SA = $L \times W \times 2 + L \times H \times 2 + W \times H \times 2$)



- A 23.13 in²
- B 23.12 in²
- C 23.11 in²
- D 23.1 in²
- E 23 in²